Global Event Query + Hard-RoI | Val | tau = 0.5

5 backbones x 4 cells; fixed epoch 4; seed 2026; original GMR parameters frozen

Backbone	Configuration	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	GMR	88.57	69.21	69.40	**7.35**	**4.65**	**10.86**	2.17	44.00	41.56
	+ Global Query	**100.00 (+11.43)**	70.59 (+1.38)	**70.71 (+1.31)**	**7.35 (+0.00)**	**4.65 (+0.00)**	**10.86 (+0.00)**	2.17 (+0.00)	48.68 (+4.68)	46.33 (+4.77)
	+ Hard-RoI	88.57 (+0.00)	69.21 (+0.00)	69.40 (+0.00)	6.71 (-0.64)	3.97 (-0.68)	9.84 (-1.02)	**2.33 (+0.16)**	44.08 (+0.08)	41.56 (+0.00)
	+ Global Query + Hard-RoI	**100.00 (+11.43)**	**70.76 (+1.55)**	**70.71 (+1.31)**	6.73 (-0.62)	3.90 (-0.75)	9.84 (-1.02)	**2.33 (+0.16)**	**48.70 (+4.70)**	**46.34 (+4.78)**
QD-DETR	GMR	80.95	**74.15**	69.11	**19.60**	**13.30**	26.09	**10.22**	45.16	40.78
	+ Global Query	92.38 (+11.43)	73.81 (-0.34)	70.16 (+1.05)	**19.60 (+0.00)**	**13.30 (+0.00)**	26.09 (+0.00)	**10.22 (+0.00)**	49.44 (+4.28)	44.73 (+3.95)
	+ Hard-RoI	80.95 (+0.00)	**74.15 (+0.00)**	69.11 (+0.00)	19.02 (-0.58)	12.47 (-0.83)	**26.23 (+0.14)**	9.48 (-0.74)	45.34 (+0.18)	40.79 (+0.01)
	+ Global Query + Hard-RoI	**93.33 (+12.38)**	73.56 (-0.59)	**70.38 (+1.27)**	18.82 (-0.78)	12.20 (-1.10)	25.64 (-0.45)	9.48 (-0.74)	**50.20 (+5.04)**	**45.17 (+4.39)**
CG-DETR	GMR	50.48	**74.74**	56.84	20.98	14.08	28.47	11.76	34.44	28.71
	+ Global Query	**97.14 (+46.66)**	73.79 (-0.95)	71.33 (+14.49)	20.98 (+0.00)	14.08 (+0.00)	28.47 (+0.00)	11.76 (+0.00)	**51.40 (+16.96)**	**46.64 (+17.93)**
	+ Hard-RoI	50.48 (+0.00)	**74.74 (+0.00)**	56.84 (+0.00)	**21.56 (+0.58)**	**14.65 (+0.57)**	**29.03 (+0.56)**	11.69 (-0.07)	34.57 (+0.13)	28.76 (+0.05)
	+ Global Query + Hard-RoI	96.67 (+46.19)	73.92 (-0.82)	**71.48 (+14.64)**	21.17 (+0.19)	14.10 (+0.02)	28.96 (+0.49)	**12.07 (+0.31)**	51.09 (+16.65)	46.51 (+17.80)
EaTR	GMR	69.05	**75.63**	65.02	**20.26**	**13.78**	**26.75**	**12.44**	41.62	36.00
	+ Global Query	**96.19 (+27.14)**	72.46 (-3.17)	**71.00 (+5.98)**	**20.26 (+0.00)**	**13.78 (+0.00)**	**26.75 (+0.00)**	**12.44 (+0.00)**	**51.17 (+9.55)**	**46.23 (+10.23)**
	+ Hard-RoI	69.05 (+0.00)	**75.63 (+0.00)**	65.02 (+0.00)	19.01 (-1.25)	13.17 (-0.61)	25.27 (-1.48)	10.91 (-1.53)	41.30 (-0.32)	35.96 (-0.04)
	+ Global Query + Hard-RoI	**96.19 (+27.14)**	72.68 (-2.95)	70.88 (+5.86)	19.53 (-0.73)	13.52 (-0.26)	26.12 (-0.63)	11.69 (-0.75)	51.06 (+9.44)	46.19 (+10.19)
FlashVTG	GMR	72.38	**74.89**	66.38	**29.16**	**18.77**	**38.28**	**14.74**	44.61	38.32
	+ Global Query	94.76 (+22.38)	72.23 (-2.66)	70.69 (+4.31)	28.96 (-0.20)	18.46 (-0.31)	**38.28 (+0.00)**	**14.74 (+0.00)**	51.50 (+6.89)	46.25 (+7.93)
	+ Hard-RoI	72.38 (+0.00)	**74.89 (+0.00)**	66.38 (+0.00)	27.08 (-2.08)	17.47 (-1.30)	36.51 (-1.77)	13.91 (-0.83)	44.40 (-0.21)	38.11 (-0.21)
	+ Global Query + Hard-RoI	**95.24 (+22.86)**	72.09 (-2.80)	**70.80 (+4.42)**	27.23 (-1.93)	17.73 (-1.04)	36.63 (-1.65)	14.24 (-0.50)	**51.84 (+7.23)**	**46.45 (+8.13)**

All values: %. Parentheses: gain versus same-backbone GMR (pp), subtracting displayed values. Bold: within-backbone best.
LoRA / Gaussian-RoI / 6D Saliency / candidate presence fusion: OFF. Baseline and Global-only are verified reused cells.
10 new Hard-only / Global+Hard runs; baseline: 0 new epochs. Standard data: 4,138 train / 465 val / 1,036 test.
Exploratory single-seed comparisons; no test checkpoint/threshold selection. Native FlashVTG score calibration and NMS retained.